Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Product-minded fullstack engineer with 8 years of experience building and shipping end-to-end features across AI-powered consumer products, healthcare platforms, and fintech systems. Proven track record of owning entire products from conception to launch, including a solo-built AI platform with multi-agent LLM pipelines, hybrid RAG search, and real-time streaming. Deep expertise in React, TypeScript, Next.js, and NodeJS, with hands-on AI integration experience using OpenAI, LangChain, vector databases, and agentic workflows. Thrives in small, fast-moving teams where shipping speed, user feedback loops, and entrepreneurial ownership define the culture.

CORE COMPETENCIES

- **Product Engineering:** End-to-end feature ownership, rapid prototyping, user-driven iteration, onboarding flows, and fullstack delivery.
- **AI & LLM Integrations:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, multi-agent systems, and agentic workflows.
- **Frontend:** React, Next.js, TypeScript, Vue.js, Redux, Zustand, Tailwind CSS, Radix UI, Recharts, and responsive UI design.
- **Backend & Data:** NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, SSE streaming, JWT authentication, and RBAC.
- **Programming Languages:** TypeScript, modern JavaScript (ESNext), NodeJS, and Python.
- **Testing & Quality:** Playwright, Jest, Cypress, CI-driven QA, Allure Reports, ESLint, and Prettier.
- **Infrastructure:** Docker, GitHub Actions, CI/CD pipelines, AWS, Vercel, Nx monorepos, and Traefik.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources, embed the widget on external sites, and query their data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, owning the feature end-to-end from architecture through launch while maintaining the legacy codebase in parallel.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and rapid iteration.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios** for asynchronous data streaming.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, delivering features end-to-end in a fast-paced startup environment.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and hands-on code reviews.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Owned full-stack integration end-to-end, connecting the **JavaScript** frontend to **NodeJS** backend services, converting business requirements into database schemas and comprehensive documentation.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. Solo-built from idea to live product serving citizens across 37 states.

- Architected and built the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (**Recharts**), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a **Remotion** video generation service.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback).
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion, with **Cohere** reranking and **Voyage AI** embeddings (708K+ chunks across 4 vector indexes), enabling citizens to query 700+ budget documents in plain English.
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats, token-budgeted context management, conversation summarization, and memory extraction. Built phone OTP and **Telegram OAuth** authentication, shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Configured **CI/CD** with **GitHub Actions**, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*